

Table 1 · Dataset, parcellations, and evaluation

Field	Value
Cohort	HCP Young Adults (FC + SC + demographics + brain-volume)
Subjects (per split)	~878 (683 train / 195 test, family-aware)
Splits	10 frozen, family-aware (siblings/twins kept on one side)
Parcellation A	Glasser — 360 regions, 64,620 edges
Parcellation B	4S456Parcels — 456 regions, 103,740 edges
Connectome rep.	upper-triangle edge vector
Primary recon method	demeaned_pearson (per-subject, train-mean removed)
Secondary metrics	avg_rank, top1_acc (fingerprint), r2/mse
Recon pipeline	PCA(256) → PLS(64) → inverse-PCA (+ BayesianRidge, KernelRidge)
Downstream estimator	BayesianRidge (reportable for scalar targets)